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METHOD AND APPARATUS FOR MANUALLY SELECTING A NETWORK

Abstract

A method comprising: determining, by a user equipment, that a non-closed access group cell is available to the user equipment in a public land mobile network, wherein the public land mobile network is identified by a public land mobile network identity and the user equipment is only allowed to access the public land mobile network via one or more closed access group cells; displaying the public land mobile network identity to a user; receiving user input selecting the public land mobile network identity; and transmitting a registration request to the public land mobile network identified by the public land mobile network identity.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. Non-Provisional patent application Ser. No. 18/089,647, filed Dec. 28, 2022 and entitled “Method and Apparatus for Manually Selecting a Network,” which is a continuation of U.S. Non-Provisional patent application Ser. No. 17/036,729, filed Sep. 29, 2020 and entitled “Method and Apparatus for Manually Selecting a Network,” which issued on Jan. 31, 2023 as U.S. Pat. No. 11,570,704, and which claims priority to, and the benefit of, United Kingdom Patent Application Serial No. 1915748.6, filed Oct. 30, 2019 and entitled “Method and Apparatus for Manually Selecting a Network,” which was published as United Kingdom Patent Publication No. GB201915748D0 on Dec. 11, 2019, the entire disclosures of each of which are hereby incorporated herein by reference in their entireties for all purposes.

TECHNICAL FIELD

[0002] The present disclosure relates to an apparatus, a method, and a computer program and in particular but not exclusively for an apparatus, methods and computer programs for manually selecting a network.

BACKGROUND

[0003] A communication system can be seen as a facility that enables communication sessions between two or more entities such as communication devices, base stations/access points and/or other nodes by providing carriers between the various entities involved in the communications path. A communication system can be provided for example by means of a communication network and one or more compatible communications devices.

[0004] Access to the communication system may be via an appropriate communications device or terminal. A communications device is provided with an appropriate signal receiving and transmitting apparatus for enabling communications, for example enabling access to a communication network or communications directly with other communications device. The communications device may access a carrier provided by a station or access point, and transmit and/or receive communications on the carrier.

[0005] The communication system and associated devices typically operate in accordance with a given standard or specification which sets out what the various entities associated with the system are permitted to do and how that should be achieved.

BRIEF SUMMARY

[0006] According to an aspect, there is provided a method comprising determining, by a user equipment, that a non-closed access group cell is available to the user equipment in a public land mobile network, wherein the public land mobile network is identified by a public land mobile network identity and the user equipment is only allowed to access the public land mobile network via one or more closed access group cells; displaying the public land mobile network identity to a user; receiving user input selecting the public land mobile network identity; and transmitting a registration request to the public land mobile network identified by the public land mobile network identity.

[0007] The public land mobile network identity may be stored in a list comprising one or more public land mobile network identities, wherein for each public land mobile network identity there is an associated zero or more closed access group identities which the user equipment is allowed to use in the public land mobile network identified by the public land mobile network identity.

[0008] For each public land mobile network identity, there may be an indication that the user equipment is only allowed to access a 5G system via closed access group cells.

[0009] The displayed public land mobile network identity may be associated with an indication that a user equipment is only allowed to access the public land mobile network identified by the public land mobile network identity via one or more closed access group cells.

[0010] The method may comprise receiving a registration accept message; and deleting the indication that the user equipment is only allowed to access a 5G system via closed access group cells for the public land mobile network identified by the public land mobile network identity.

[0011] The method may comprise receiving a registration reject message; and retaining the indication that the user equipment is only allowed to access 5G system via closed access group cells.

[0012] Displaying the public land mobile network identity may comprise displaying the public land mobile network identity associated with a list of one or more closed access group identities, wherein there exists a closed access group cell available to a user equipment in the public land mobile network identified by the public land mobile network identity, and wherein for each of the closed access group identities, it is indicated whether the user equipment is allowed for the closed access group identity.

[0013] The list comprising the one or more closed access group identities may include a closed access group identity field, which is either populated with a respective closed access group identity if the entry has a closed access group identity, or is populated with an entry indicating a null closed access group identity if the entry does not have a closed access group identity.

[0014] The method may comprise displaying the public land mobile network identity without a list of one or more closed access group identities in addition to the public land mobile network identity displayed with a list of one or more closed access group identities.

[0015] According to an aspect, there is provided an apparatus comprising means for determining that a non-closed access group cell is available to the user equipment in a public land mobile network, wherein the public land mobile network is identified by a public land mobile network identity and the user equipment is only allowed to access the public land mobile network via one or more closed access group cells; displaying the public land mobile network identity to a user; receiving user input selecting the public land mobile network identity; and transmitting a registration request to the public land mobile network identified by the public land mobile network identity.

[0016] The public land mobile network identity may be stored in a list comprising one or more public land mobile network identities, wherein for each public land mobile network identity there is an associated zero or more closed access group identities which the user equipment is allowed to use in the public land mobile network identified by the public land mobile network identity.

[0017] For each public land mobile network identity, there may be an indication that the user equipment is only allowed to access a 5G system via closed access group cells.

[0018] The displayed public land mobile network identity may be associated with an indication that a user equipment is only allowed to access the public land mobile network identified by the public land mobile network identity via one or more closed access group cells.

[0019] The means may be further configured to perform receiving a registration accept message; and deleting the indication that the user equipment is only allowed to access a 5G system via closed access group cells for the public land mobile network identified by the public land mobile network identity.

[0020] The means may be further configured to perform receiving a registration reject message;

and retaining the indication that the user equipment is only allowed to access 5G system via closed access group cells.

[0021] Displaying the public land mobile network identity may comprise displaying the public land mobile network identity associated with a list of one or more closed access group identities, wherein there exists a closed access group cell available to a user equipment in the public land mobile network identified by the public land mobile network identity, and wherein for each of the closed access group identities, it is indicated whether the user equipment is allowed for the closed access group identity.

[0022] The list comprising the one or more closed access group identities may include a closed access group identity field, which is either populated with a respective closed access group identity if the entry has a closed access group identity, or is populated with an entry indicating a null closed access group identity if the entry does not have a closed access group identity.

[0023] The means may be further configured to perform displaying the public land mobile network identity without a list of one or more closed access group identities in addition to the public land mobile network identity displayed with a list of one or more closed access group identities.

[0024] According to an aspect, there is provided an apparatus comprising at least one processor; and at least one memory including computer program code, the at least one memory and computer program code configured to, with the at least one processor, cause the apparatus to perform: determining that a non-closed access group cell is available to a user equipment in a public land mobile network, wherein the public land mobile network is identified by a public land mobile network identity and the user equipment is only allowed to access the public land mobile network via one or more closed access group cells; displaying the public land mobile network identity to a user; receiving user input selecting the public land mobile network identity; and transmitting a registration request to the public land mobile network identified by the public land mobile network identity.

[0025] The public land mobile network identity may be stored in a list comprising one or more public land mobile network identities, wherein for each public land mobile network identity there is an associated zero or more closed access group identities which the user equipment is allowed to use in the public land mobile network identified by the public land mobile network identity.

[0026] For each public land mobile network identity there may be an indication that the user equipment is only allowed to access a 5G system via closed access group cells.

[0027] The displayed public land mobile network identity may be associated with an indication that a user equipment is only allowed to access the public land mobile network identified by the public land mobile network identity via one or more closed access group cells.

[0028] The apparatus may be further configured to perform: receiving a registration accept message; and deleting the indication that the user equipment is only allowed to access a 5G system via closed access group cells for the public land mobile network identified by the public land mobile network identity.

[0029] The apparatus may be further configured to perform: receiving a registration reject message; and retaining the indication that the user equipment is only allowed to access 5G system via closed access group cells.

[0030] Displaying the public land mobile network identity may comprise displaying the public land mobile network identity associated with a list of one or more closed access group identities, wherein there exists a closed access group cell available to a user equipment in the public land mobile network identified by the public land mobile network identity, and wherein for each of the closed access group identities, it is indicated whether the user equipment is allowed for the closed access group identity.

[0031] The list comprising the one or more closed access group identities may include a closed access group identity field, which is either populated with a respective closed access group identity if the entry has a closed access group identity, or is populated with an entry indicating a null closed

access group identity if the entry does not have a closed access group identity.

[0032] The apparatus may be further configured to perform: displaying the public land mobile network identity without a list of one or more closed access group identities in addition to the public land mobile network identity displayed with a list of one or more closed access group identities.

[0033] According to an aspect, there is provided a non-transitory computer readable storage medium comprising computer readable instructions, which when run by at least one processor, cause an apparatus to determine that a non-closed access group cell is available to the user equipment in a public land mobile network, wherein the public land mobile network is identified by a public land mobile network identity and the user equipment is only allowed to access the public land mobile network via one or more closed access group cells; display the public land mobile network identity to a user; receive user input selecting the public land mobile network identity; and transmit a registration request to the public land mobile network identified by the public land mobile network identity.

[0034] According to an aspect, there is provided a computer program comprising instructions for causing an apparatus to perform at least determining that a non-closed access group cell is available to the user equipment in a public land mobile network, wherein the public land mobile network is identified by a public land mobile network identity and the user equipment is only allowed to access the public land mobile network via one or more closed access group cells; displaying the public land mobile network identity to a user; receiving user input selecting the public land mobile network identity; and transmitting a registration request to the public land mobile network identified by the public land mobile network identity.

[0035] According to an aspect, there is provided an apparatus comprising: determining circuitry for determining that a non-closed access group cell is available to the user equipment in a public land mobile network, wherein the public land mobile network is identified by a public land mobile network identity and the user equipment is only allowed to access the public land mobile network via one or more closed access group cells; displaying circuitry for displaying the public land mobile network identity to a user; receiving circuitry for receiving user input selecting the public land mobile network identity; and transmitting circuitry for transmitting a registration request to the public land mobile network identified by the public land mobile network identity.

[0036] According to another aspect, a method can be carried out by means, such as a computing device or other apparatus, based on computer-executable instructions, such as instructions stored as program codes. The method can include: determining, by a user equipment, that a cell is available to the user equipment in a public land mobile network, wherein the cell is not a closed access group cell for the public land mobile network; providing, on the user equipment, at least the public land mobile network for user selection; determining, by the user equipment, that a public land mobile network identity identifying the public land mobile network is associated with an indication that the user equipment is only allowed to access a communication system via closed access group cells; indicating, on the user equipment, that the user equipment is only allowed to access the public land mobile network via one or more closed access group cells; receiving a user selection of the public land mobile network; and, based on the received user selection, transmitting a registration request to the selected public land mobile network.

[0037] In some embodiments, the public land mobile network identity is stored in a list comprising one or more public land mobile network identities, wherein for each public land mobile network identity there is an associated zero or more closed access group identities which the user equipment is allowed to use in the public land mobile network identified by the public land mobile network identity. In some embodiments, for each public land mobile network identity there is an indication that the user equipment is only allowed to access a 5G system via closed access group cells. In some embodiments, the method can further include: receiving a registration accept message; and deleting the indication that the user equipment is only allowed to access a 5G system via closed

access group cells for the public land mobile network identified by the public land mobile network identity. In some embodiments, the method can further include: receiving a registration reject message; and retaining the indication that the user equipment is only allowed to access 5G system via closed access group cells. In some embodiments, providing the public land mobile network comprises displaying the public land mobile network identity associated with a list of one or more closed access group identities, wherein there exists a closed access group cell available to a user equipment in the public land mobile network identified by the public land mobile network identity, and wherein for each of the closed access group identities, it is indicated whether the user equipment is allowed for the closed access group identity.

[0038] In some embodiments, the list can include the one or more closed access group identities includes a closed access group identity field, which is either populated with a respective closed access group identity if the entry has a closed access group identity, or is populated with an entry indicating a null closed access group identity if the entry does not have a closed access group identity. In some embodiments, the method can further include: displaying the public land mobile network identity without a list of one or more closed access group identities in addition to the public land mobile network identity displayed with a list of one or more closed access group identities.

[0039] According to another aspect, an apparatus can be provided that includes: at least one processor; and at least one memory including computer programs code, the at least one memory and computer program code being configured to, with the at least one processor, cause the apparatus at least to: determine that a cell is available to the apparatus in a public land mobile network, wherein the cell is not a closed access group cell for the public land mobile network; provide at least the public land mobile network for user selection; determine that a public land mobile network identity identifying the public land mobile network is associated with an indication that the apparatus is only allowed to access a communication system via closed access group cells; indicate that the apparatus is only allowed to access the public land mobile network via one or more closed access group cells; receive a user selection of the public land mobile network; and, based on the received user selection, transmit a registration request to the selected public land mobile network.

[0040] In some embodiments, the public land mobile network identity is stored in a list comprising one or more public land mobile network identities, wherein for each public land mobile network identity there is an associated zero or more closed access group identities which the apparatus is allowed to use in the public land mobile network identified by the public land mobile network identity. In some embodiments, for each public land mobile network identity there is an indication that the apparatus is only allowed to access a 5G system via closed access group cells. In some embodiments, the at least one memory and the computer program codes are further configured to, with the at least one processor, cause the apparatus at least to: receive a registration accept message; and delete the indication that the apparatus is only allowed to access a 5G system via closed access group cells for the public land mobile network identified by the public land mobile network identity. In some embodiments, the at least one memory and the computer program codes are further configured to, with the at least one processor, cause the apparatus at least to: receive a registration reject message; and retain the indication that the apparatus is only allowed to access 5G system via closed access group cells.

[0041] In some embodiments, providing the public land mobile network comprises displaying the public land mobile network identity associated with a list of one or more closed access group identities, wherein there exists a closed access group cell available to a apparatus in the public land mobile network identified by the public land mobile network identity, and wherein for each of the closed access group identities, it is indicated whether the apparatus is allowed for the closed access group identity. In some embodiments, the list can include the one or more closed access group identities includes a closed access group identity field, which is either populated with a respective

closed access group identity if the entry has a closed access group identity, or is populated with an entry indicating a null closed access group identity if the entry does not have a closed access group identity. In some embodiments, the at least one memory and the computer program codes are further configured to, with the at least one processor, cause the apparatus at least to: display the public land mobile network identity without a list of one or more closed access group identities in addition to the public land mobile network identity displayed with a list of one or more closed access group identities.

[0042] According to another aspect, a computer program can be provided, such as a computer program embodied on a non-transitory computer-readable medium. In some embodiments, the computer program can include computer-executable codes which, when executed in hardware, cause the hardware to perform at least: determining, by a user equipment, that a cell is available to the user equipment in a public land mobile network, wherein the cell is not a closed access group cell for the public land mobile network; providing at least the public land mobile network for user selection; determining, by the user equipment, that a public land mobile network identity identifying the public land mobile network is associated with an indication that the user equipment is only allowed to access a communication system via closed access group cells; indicating, on the user equipment, that the user equipment is only allowed to access the public land mobile network via one or more closed access group cells; receiving a user selection of the public land mobile network; and, based on the received user selection, transmitting a registration request to the selected public land mobile network.

[0043] In some embodiments, the computer-executable code, when executed in the hardware, can further cause the hardware to perform: storing the public land mobile network identity in a list comprising one or more public land mobile network identities. In some embodiments, for each public land mobile network identity there is an associated zero or more closed access group identities which the user equipment is allowed to use in the public land mobile network identified by the public land mobile network identity. In some embodiments, for each public land mobile network identity there is an indication that the user equipment is only allowed to access a 5G system via closed access group cells. In some embodiments, the computer-executable code, when executed in the hardware, can further cause the hardware to perform: receiving a registration accept message; and deleting the indication that the user equipment is only allowed to access a 5G system via closed access group cells for the public land mobile network identified by the public land mobile network identity.

[0044] In the above, many different aspects have been described. It should be appreciated that further aspects may be provided by the combination of any two or more of the aspects described above.

[0045] Various other aspects are also described in the following detailed description and in the attached claims.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0046] Some example embodiments will now be described in further detail, by way of example only, with reference to the following examples and accompanying drawings, in which:

[0047] FIG. 1 shows a representation of a network system according to some example embodiments;

[0048] FIG. 2 shows a representation of a control apparatus according to some example embodiments;

[0049] FIG. 3 shows a representation of an apparatus according to some example embodiments;

[0050] FIG. 4 shows a representation of a registration procedure according to some example

embodiments;

[0051] FIG. 5 shows a representation of non-volatile memory media according to some example embodiments; and

[0052] FIG. 6 shows a method according to some example embodiments.

DETAILED DESCRIPTION

[0053] In the following certain embodiments are explained with reference to mobile communication devices capable of communication via a wireless cellular system and mobile communication systems serving such mobile communication devices. Before explaining in detail the exemplifying embodiments, certain general principles of a wireless communication system, access systems thereof, and mobile communication devices are briefly explained with reference to FIGS. 1, 2 and 3 to assist in understanding the technology underlying the described examples.

[0054] FIG. 1 shows a schematic representation of a 5G system (5GS). The 5GS may be comprised by a terminal or user equipment (UE), a 5G radio access network (5GRAN) or next generation radio access network (NG-RAN), a 5G core network (5GC), one or more application function (AF) and one or more data networks (DN).

[0055] The 5G-RAN may comprise one or more gNodeB (GNB) or one or more gNodeB (GNB) distributed unit functions connected to one or more gNodeB (GNB) centralized unit functions.

[0056] The 5GC may comprise the following entities: Network Slice Selection Function (NSSF); Network Exposure Function; Network Repository Function (NRF); Policy Control Function (PCF); Unified Data Management (UDM); Application Function (AF); Authentication Server Function (AUSF); an Access and Mobility Management Function (AMF); and Session Management Function (SMF).

[0057] FIG. 2 illustrates an example of a control apparatus **200** for controlling a function of the 5GRAN or the 5GC as illustrated on FIG. 1. The control apparatus may comprise at least one random access memory (RAM) **211a**, at least on read only memory (ROM) **211b**, at least one processor **212**, **213** and an input/output interface **214**. The at least one processor **212**, **213** may be coupled to the RAM **211a** and the ROM **211b**. The at least one processor **212**, **213** may be configured to execute an appropriate software code **215**. The software code **215** may for example allow to perform one or more steps to perform one or more of the present aspects. The software code **215** may be stored in the ROM **211b**. The control apparatus **200** may be interconnected with another control apparatus **200** controlling another function of the 5GRAN or the 5GC. In some embodiments, each function of the 5GRAN or the 5GC comprises a control apparatus **200**. In alternative embodiments, two or more functions of the 5GRAN or the 5GC may share a control apparatus.

[0058] FIG. 3 illustrates an example of a terminal **300**, such as the terminal illustrated on FIG. 1. The terminal **300** may be provided by any device capable of sending and receiving radio signals. Non-limiting examples comprise a user equipment, a mobile station (MS) or mobile device such as a mobile phone or what is known as a 'smart phone', a computer provided with a wireless interface card or other wireless interface facility (e.g., USB dongle), a personal data assistant (PDA) or a tablet provided with wireless communication capabilities, a machine-type communications (MTC) device, an Internet of things (IoT) type communication device or any combinations of these or the like. The terminal **300** may provide, for example, communication of data for carrying communications. The communications may be one or more of voice, electronic mail (email), text message, multimedia, data, machine data and so on.

[0059] The terminal **300** may receive signals over an air or radio interface **307** via appropriate apparatus for receiving and may transmit signals via appropriate apparatus for transmitting radio signals. In FIG. 3 transceiver apparatus is designated schematically by block **306**. The transceiver apparatus **306** may be provided for example by means of a radio part and associated antenna arrangement. The antenna arrangement may be arranged internally or externally to the mobile device.

[0060] The terminal **300** may be provided with at least one processor **301**, at least one memory ROM **302a**, at least one RAM **302b** and other possible components **303** for use in software and hardware aided execution of tasks it is designed to perform, including control of access to and communications with access systems and other communication devices. The at least one processor **301** is coupled to the RAM **311a** and the ROM **311b**. The at least one processor **301** may be configured to execute an appropriate software code **308**. The software code **308** may for example allow to perform one or more of the present aspects. The software code **308** may be stored in the ROM **311b**.

[0061] The processor, storage and other relevant control apparatus can be provided on an appropriate circuit board and/or in chipsets. This feature is denoted by reference **304**. The device may optionally have a user interface such as a keypad **305**, touch sensitive screen or pad, combinations thereof or the like. Optionally one or more of a display, a speaker and a microphone may be provided depending on the type of the device.

[0062] Certain cells in a wireless communications network may be designated as closed access group (CAG) cells. A CAG defines a limited group of UEs that may access a particular CAG cell or group of cells. UEs outside the CAG may not be able to access the cell(s) in the CAG.

[0063] In some cases, manual CAG selection may be permitted. In manual CAG selection, a user may be allowed to manually choose a CAG ID that is not in an allowed CAG list. In order to make this functionality available, the UE may display one or more CAG IDs that are not in the allowed CAG list. The UE may initiate a registration procedure if the user selects a non-allowed CAG ID. If the registration procedure is successful, the UE may add the selected CAG ID to the allowed CAG list.

[0064] In some cases, the UE may only be allowed to access a public land mobile network (PLMN) via certain CAG cells. However, a user may select a PLMN that is only allowed be accessed via a CAG that is not in the allowed CAG list. As such, the user may have to be able to select a PLMN without selecting a CAG, even if the UE is only allowed to access the PLMN via CAG cells.

[0065] In some example embodiments, there is provided a method as shown in FIG. **6**.

[0066] At step **600**, the method comprises determining, by a user equipment, that a non-closed access group cell is available to the user equipment in a public land mobile network. The public land mobile network is identified by a public land mobile network identity and the user equipment is only allowed to access the public land mobile network via one or more closed access group cells.

[0067] At step **602**, the method comprises displaying the public land mobile network identity to a user.

[0068] At step **604**, the method comprises receiving user input selecting the public land mobile network identity.

[0069] At step **606**, the method comprises transmitting a registration request to the public land mobile network identified by the public land mobile network identity.

[0070] An example of a registration procedure according to some example embodiments is shown in FIG. **4**.

[0071] As shown in FIG. **4**, the user equipment transmits, to an access management function (AMF), a message comprising a registration request.

[0072] When the registration procedure is successful, the AMF transmits to the user equipment a registration accept message. Or if the registration procedure is not successful, for example if the AMF needs to reject the registration request from the UE, the AMF sends a registration reject message.

[0073] In some example embodiments, the displayed information indicating one or more public land mobile networks may indicate which public land mobile networks are available using all supported radio access technologies. PLMNs meeting the criteria above may presented in the following order: [0074] i) if one or more equivalent home public land mobile networks (EHPLMNs) are available, then based on an optional data field on a universal mobile

telecommunications system subscriber identification module (USIM) either only the highest priority available EHPLMN is to be presented to the user, or all available EHPLMNs are presented to the user in priority order. Alternatively, if an EHPLMN list is not present or is empty, or if the data field is not present on the USIM, the home public land mobile network (HPLMN) is presented instead of the EHPLMN. [0075] ii) PLMN/radio access technology combinations contained in the USIM, in priority order. The PLMN/radio access technologies may be stored in a “User Controlled PLMN Selector with Access Technology” data file in the USIM; [0076] iii) PLMN/radio access technology combinations contained in the USIM, in priority order. The PLMN/radio access technology combinations may be stored in an “Operator Controlled PLMN Selector with Access Technology” data file in the USIM. Additionally or alternatively, the PLMN/radio access technology combinations may be stored in a mobility engine (ME); [0077] iv) other PLMN/access technology combinations with received a highest quality signal. If more than one PLMN/radio access technology have the same highest signal quality, the more than one PLMN/radio access technologies with the highest signal quality may be displayed in a random order; and [0078] v) other PLMN/radio access technology combinations in order of decreasing signal quality.

[0079] In some example embodiments, if the UE supports the CAG feature, and the allowed CAG list comprises at least one entry, then for each PLMN/radio access technology combination of NG-RAN access technology, the UE may display the PLMN/radio access technology combination and a list of one or more CAG identifiers (IDs), such that for each CAG-ID there is an available CAG cell which broadcasts the CAG-ID for the PLMN. In addition, if there exists an entry with the PLMN ID of the PLMN in a “CAG information list” and the CAG-ID is included in the allowed CAG list, the UE may display an indication that the CAG-ID is allowed. Additionally or alternatively, the UE may display whether the CAG-ID is allowed whenever a CAG-ID is displayed.

[0080] Alternatively, in some example embodiments the UE may display the PLMN/radio access technology combination without a list of CAG-IDs, if there is an available cell which is not a CAG cell for the PLMN. In addition, if there exists an entry with the PLMN ID of the PLMN in the “CAG information list” including an indication that the UE is only allowed to access 5GS via CAG cells, the UE may present to the user an indication that the UE is only allowed to access 5GS (or the PLMN) via CAG cells. Additionally or alternatively, the UE may display whether the UE is only allowed to access the PLMN via CAG cells whenever a PLMN/radio access technology combination is displayed without a list of CAG-IDs.

[0081] In some example embodiments, if there exists both an available CAG cell and an available non-CAG NG-RAN cell for the PLMN, the PLMN/radio access technology combination may be displayed twice: firstly a list of PLMN/radio access technology combinations with CAG-IDs and the secondly a list of PLMN/radio access technology combinations without CAG-IDs.

[0082] Alternatively, in some example embodiments the UE may display a list of PLMN/access technology combinations once. The list may include a CAG-ID field, which is either populated with the CAG ID if an entry has a CAG ID, or is populated with an entry indicating a null CAG-ID if the entry does not have a CAG ID.

[0083] In some example embodiments, how the UE displays the PLMN/radio access technology combination and a list of CAG-IDs, such that there exists both an available CAG cell and an available non-CAG NG-RAN cell for the PLMN, may be UE implementation specific.

[0084] In some example embodiments, the UE may initiate the registration procedure by sending a REGISTRATION REQUEST message to the AMF when either: [0085] (a) the UE has selected a CAG-ID which is not included in the allowed CAG list for the selected PLMN; or [0086] (b) a CAG-ID in a PLMN for which the entry in the “CAG information list” does not exist; or [0087] (c) the UE has selected, without selecting a CAG-ID, a PLMN for which the entry in the “CAG information list” includes an indication that the UE is only allowed to access 5GS via CAG cells.

[0088] If the UE has initiated the registration procedure due to manual CAG selection and receives

a REGISTRATION ACCEPT message and the UE has initiated the registration procedure in a CAG cell, then the UE may add the selected CAG-ID in the allowed CAG list for the current PLMN (if not already included).

[0089] Alternatively, if the UE has initiated the registration procedure due to manual CAG selection and receives a REGISTRATION ACCEPT message and the UE has initiated the registration procedure in a non-CAG cell, then the UE may delete an indication that the UE is only allowed to access 5GS via CAG cells” from the entry of the “CAG information list” for the current PLMN (if any).

[0090] In some example embodiments, the AMF may reject a registration request from the UE if the UE does not pass CAG access control. In this case, the AMF can send a REGISTRATION REJECT message and “CAG information list” or part of the “CAG information list”. For example, if the UE accesses a PLMN with selecting a CAG-ID which the UE is not allowed to use in the PLMN, the network can send the allowed CAG list for the PLMN together with the REGISTRATION REJECT message.

[0091] FIG. 5 shows a schematic representation of non-volatile memory media **500a** (e.g., computer disc (CD) or digital versatile disc (DVD)) and **500b** (e.g., universal serial bus (USB) memory stick) storing instructions and/or parameters **502** which when executed by a processor allow the processor to perform one or more of the steps of the methods described previously.

[0092] It is to be noted that embodiments of the present invention may be implemented as circuitry, in software, hardware, application logic or a combination of software, hardware and application logic. In an example embodiment, the application logic, software or an instruction set is maintained on any one of various conventional computer-readable media. In the context of this document, a “computer-readable medium” may be any media or means that can contain, store, communicate, propagate or transport the instructions for use by or in connection with an instruction execution system, apparatus, or device, such as a computer or smart phone, or user equipment.

[0093] As used in this application, the term “circuitry” refers to all of the following: (a) hardware-only circuit implementations (such as implementations in only analog and/or digital circuitry) and (b) to combinations of circuits and software (and/or firmware), such as (as applicable): (i) to a combination of processor(s) or (ii) to portions of processor(s)/software (including digital signal processor(s)), software, and memory(ies) that work together to cause an apparatus, such as a mobile phone or server, to perform various functions) and (c) to circuits, such as a microprocessor(s) or a portion of a microprocessor(s), that require software or firmware for operation, even if the software or firmware is not physically present. This definition of ‘circuitry’ applies to all uses of this term in this application, including in any claims. As a further example, as used in this application, the term “circuitry” would also cover an implementation of merely a processor (or multiple processors) or portion of a processor and its (or their) accompanying software and/or firmware. The term “circuitry” would also cover, for example and if applicable to the particular claim element, a baseband integrated circuit or applications processor integrated circuit for a mobile phone or a similar integrated circuit in server, a cellular network device, or other network device.

[0094] The described features, advantages, and characteristics of the invention can be combined in any suitable manner in one or more embodiments. One skilled in the relevant art will recognize that the invention can be practiced without one or more of the specific features or advantages of a particular embodiment. In other instances, additional features and advantages can be recognized in certain embodiments that may not be present in all embodiments of the invention. One having ordinary skill in the art will readily understand that the invention as discussed above may be practiced with steps in a different order, and/or with hardware elements in configurations which are different than those which are disclosed. Therefore, although the invention has been described based upon these preferred embodiments, it would be apparent to those of skill in the art that

certain modifications, variations, and alternative constructions would be apparent, while remaining within the spirit and scope of the invention.

Claims

- 1.** A method comprising: determining, by a user equipment, that a cell is available to the user equipment in a public land mobile network and that the cell is not a closed access group cell for the public land mobile network, wherein the user equipment is only allowed to access the public land mobile network via one or more closed access group cells; based on determining that the cell is available in the public land mobile network and that the cell is not a closed access group cell for the public land mobile network, displaying, by the user equipment, at least the public land mobile network for user selection; receiving, at the user equipment, a user selection of the public land mobile network displayed via the user equipment; and based on the user selection of the public land mobile network displayed via the user equipment, transmitting, via the user equipment, a registration request to the selected public land mobile network.
- 2.** The method of claim 1, wherein the displaying at least the public land mobile network further comprises: displaying, by the user equipment, the public land mobile network without a list of closed access group identities.
- 3.** The method of claim 2, further comprising: displaying, by the user equipment, the public land mobile network and a list of one or more closed access group identities, wherein, for each closed access group identity, there is an available closed access group cell which broadcasts the closed access group identity in the public land mobile network.
- 4.** The method of claim 3, wherein the closed access group identity of the closed access group cell is included in a list of closed access group identities which the user equipment is allowed to use in the public land mobile network.
- 5.** The method of claim 3, wherein, for each of the closed access group identities, it is indicated whether the user equipment is allowed for the closed access group identity.
- 6.** The method of claim 1, wherein an entry for the public land mobile network in a closed access group information list includes an indication that the user equipment is only allowed to access the public land mobile network via one or more closed access group cells.
- 7.** The method of claim 6, wherein a public land mobile network identity of the public land mobile network is stored in the closed access group information list, wherein the closed access group information list comprises one or more public land mobile network identities, and wherein, for each public land mobile network identity, there is an associated zero or more closed access group identities which the user equipment is allowed to use in the public land mobile network.
- 8.** The method of claim 7, wherein, for each public land mobile network identity, there is an indication that the user equipment is only allowed to access a fifth generation (5G) system via closed access group cells.
- 9.** The method of claim 8, further comprising: receiving, at the user equipment, a registration accept message; and deleting, on the user equipment, the indication that the user equipment is only allowed to access a 5G system via closed access group cells for the public land mobile network.
- 10.** The method of claim 8, further comprising: receiving, at the user equipment, a registration reject message; and retaining, on the user equipment, the indication that the user equipment is only allowed to access a 5G system via closed access group cells.
- 11.** An apparatus comprising: at least one processor; and at least one memory comprising instructions stored thereon that, when executed by the at least one processor, cause the apparatus to perform at least: determining that a cell is available to the apparatus in a public land mobile network and that the cell is not a closed access group cell for the public land mobile network, wherein the apparatus is only allowed to access the public land mobile network via one or more closed access group cells; based on determining that the cell is available in the public land mobile

network and that the cell is not a closed access group cell for the public land mobile network, displaying for user selection via the apparatus at least the public land mobile network; receiving a user selection of the public land mobile network displayed via the apparatus; and based on the user selection of the public land mobile network displayed via the apparatus, transmitting a registration request to the selected public land mobile network.

12. The apparatus of claim 11, wherein the instructions stored on the at least one memory, when executed by the at least one processor, further cause the apparatus to perform: displaying the public land mobile network without a list of closed access group identities.

13. The apparatus of claim 12, wherein the instructions stored on the at least one memory, when executed by the at least one processor, further cause the apparatus to perform: displaying the public land mobile network and a list of one or more closed access group identities, wherein for each closed access group identity there is an available closed access group cell which broadcasts the closed access group identity in the public land mobile network.

14. The apparatus of claim 13, wherein the closed access group identity of the closed access group cell is included in a list of closed access group identities which the apparatus is allowed to use in the public land mobile network.

15. The apparatus of claim 13, wherein, for each of the closed access group identities, it is indicated whether the apparatus is allowed for the closed access group identity.

16. The apparatus of claim 13, wherein the list of one or more closed access group identities includes a closed access group identity field, which is populated with either: a respective closed access group identity if the entry has a closed access group identity or an entry indicating a null closed access group identity if the entry does not have a closed access group identity.

17. The apparatus of claim 11, wherein an entry for the public land mobile network in a closed access group information list includes an indication that the apparatus is only allowed to access the public land mobile network via one or more closed access group cells.

18. The apparatus of claim 17, wherein a public land mobile network identity of the public land mobile network is stored in the closed access group information list, wherein the closed access group information list comprises one or more public land mobile network identities, wherein, for each public land mobile network identity, there is an associated zero or more closed access group identities which the apparatus is allowed to use in the public land mobile network.

19. The apparatus of claim 18, wherein, for each public land mobile network identity, there is an indication that the apparatus is only allowed to access a fifth generation (5G) system via closed access group cells.

20. The apparatus of claim 19, wherein the apparatus is further configured to perform: receiving a registration accept message; and deleting the indication that the apparatus is only allowed to access a 5G system via closed access group cells for the public land mobile network.

21. The apparatus of claim 19, wherein the apparatus is further configured to perform: receiving a registration reject message; and retaining the indication that the apparatus is only allowed to access a 5G system via closed access group cells.

22. A non-transitory computer-readable storage medium comprising program codes stored thereon that, when executed by at least one processor of an apparatus, cause the apparatus to perform at least: determining that a cell is available to the apparatus in a public land mobile network and that the cell is not a closed access group cell for the public land mobile network, wherein the apparatus is only allowed to access the public land mobile network via one or more closed access group cells; based on determining that the cell is available in the public land mobile network and that the cell is not a closed access group cell for the public land mobile network, displaying for user selection via the apparatus at least the public land mobile network; receiving a user selection of the public land mobile network displayed via the apparatus; and based on the user selection of the public land mobile network displayed via the apparatus, transmitting a registration request to the selected public land mobile network.

